

ECSP3000 Project Brief

Designing and Implementing an Automated & Reproducible Testbed for Linux Rootkit Detectors, Aiming to Identify and Implement a Novel Rootkit/Detector Technique

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September 27, 2026

1. Project Aims

Current research into the evaluation of rootkits and rootkit detectors rely on bespoke evaluation methods which cannot be robustly reproduced by third partys.

This tool aims to provide developers, researchers and system administrators with an automated and reproducible testing framework for rootkits & rootkit detectors on the modern Linux kernel to foster future development and testing and to provide greater transparency on what rootkit techniques remain evasive to standard detection tools.

As an extension, I will use this tool to identify and evaluate current avenues in the modern (7.2) Linux Kernel for rootkit persistence and design, finishing with an evaluation using the aforementioned benchmarking tool.

2. Project Objectives

This project will have the following aims which will be used to gauge success:

1. To have a testing framework which, upon specifying a set of rootkits & rootkit detectors, provides timely & accurate results which can then be reproduced exactly upon rerun.

2. The testing framework allows developers to easily input their own rootkit detectors and new rootkit PoCs to allow for future testing.
3. (Stretch) To research and develop a rootkit PoC for the modern Linux Kernel, following with an evaluation using this testing framework.
4. Using a dataset of results produced by the framework or, if successful, the aforementioned PoC, evaluate the current landscape of on-host rootkit detection on the modern Linux Kernel.

3. Limitations & Scope

In order to scope my project appropriately, here are a list of things that I currently plan on **not** doing.

1. My testing framework and therefore my rootkit research will only be concerned with Kernel Space Rootkits. This excludes user-space, firmware or any other rootkit classes from my research.
2. All detector techniques compatible with the tool must run **on host**. This rules out from my research any hypervisor-based techniques or methods such as off-host memory inspection as detection techniques.
3. Testing rootkits that implement anti-sandbox measures fall out of scope for this project due to the virtualised environment employed by the testing framework.